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TAFILA TECHNICAL UNIVERSITY
Faculty of Engineering
Department of Computer and Communications Engineering

Microprocessor and Assembly Language/ Microprocessor 1

First Exam

18/25

Student Name: [redacted] Time allowed: 60 minutes

6

Question#1

Select the correct answer for each of the following (**6 pts**): BL = BL + BH

[1] Which instruction among the following is Valid?

a) PUSH BL ✗

b) ADD BL, BH

c) MOV BL, 101010101B ✗

d) MOV BL, CX ✗ BL = CX

[2] How many address bits are required to address/index **2Mbytes** of memory? $2^1 \; 2^{20} = 2^{21}$

a) 20 b) 21 c) 22 d) 24

[3] What is the address of the last memory location in a 2Kbytes memory chip? 1K = 400h 2K = 800h

a) 1sFF b) 3FF c) 7FF d) FFF 7FF

[4] Splitting the internal structure of the 8086 microprocessor into two sections (the **BIU** and the **EU**) makes the CPU process information faster by:

a) Verifying the pipelining concept

b) Increasing the address bus

c) Performing two loads from memory at the same time

d) Increasing the data bus

400 0 - 3FF 800

[5] Which one among the 8086 microprocessor resources is used to store fetched instructions?

a) Registers b) Instruction Decoder

c) ALU d) Instruction Queue

8086/8088
16

[6] What is the size of the data bus in the 8088 microprocessor?

a) 8 b) 16 c) 24 d) 32

8086 →
داخل 16 → 16 خارج

8088
داخل 16 → 8 خارج

6

Question#2

Given the following 8086 assembly and machine code which is stored in the code segment, answer the followed questions? (**7 pts**)

Logical Address	Machine Code	Assembly Instructions
54B0:0346	B078	MOV AL,78H
54B0:0348	B426	MOV AH,26H
54B0:034A	02E0	ADD AH, AL

[1] Determine the physical addresses of the two memory locations that hold the third instruction?

54B00 + 034A = ⟹ 54B00 + 034A = 54E4A

= 54E4A

1

[2] What is the value of IP register needed to fetch the second instruction?

0348

1

[3] What would be the content of AX register after executing the whole code?

AL = 78H
AH = 26H
AH = AH + AL

AH = 26 + 78 ⟹ 26 + 78 = 9E
AH = 9E
AX = 9E78

2

15 +
26 -
26 - 16 =

[4] Determine the physical addresses of the last two locations in Code Segment?

لو طالب uppe و يلي قبل upper

upper ⟹ 54B00 + FFFF = ⟹ 54B00 + FFFF = 64AFF

= 64AFF

before upper ⟹ 54B00 + FFFE = ⟹ 54B00 + FFFE = 64AFE

= 64AFE

2

⟶ or from upper we will know the before

3

Question#3

Given that the register values of an 8086 microprocessor are: (6 pts)

CS : 43A0H	SP : 0B2BH
DS : 22A0H	BP : 0269H
SS : 6F20H	DI : 4598H
SI : 02BCH	

[1] Indicate the PHYSICAL addresses accessed by the following instructions, if you believe the instruction is invalid for any reason indicate that there is an ERROR in the instruction. Assume that all instructions take place with the above register contents, and the actions of instruction A will not affect the registers seen by subsequent instructions.

a) PUSH AX

SS:SP => 6F20 : 0B2B

push1 => 6F200 + 0B2B = => 6F200 + 0B2B = 6FD2B

phy1 => = 6FD2B

SS:SP-1 => 6F20 : 0B2A

push2 => 6F200 + 0B2A = => 6F200 + 0B2A = 6FD2A

b) MOV AL, [200] phy2 => = 6FD2A

ERROR

[2] Determine the value of the BX register that loads the content of memory location 23268H using the instruction MOV AL, [BX]?

23268 - 22A00 = 0868

phys = DS X 10 + offest of [BX]

23268 = 22A00 + offest of [BX]

offest of [BX] = 23268 - 22A00

offeso f [BX] = 0868

2

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Tafila Technical University
Faculty of Engineering
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Microprocessor and Assembly Language
Fall 2022 – 2nd Exam

Name: [redacted] Time Allowed: 60 minutes

11/25

6

Question#1

Choose the correct answer for each of the following (**9 points**):

1] Which register value is true after executing the following three instructions?

MOV AX, 21
MOV BH, 5
DIV BH

a) AL= 01 b) AH= 04 c) AX= 0104H d) AX= 0004

2] What would be the content of memory location 103 after defining the following?

ORG 100H
DATA1 DQ 135792468H

a) 35H b) 46H c) 92H d) 79H

3] What is the content of the AL register after executing the following?

MOV AL, 2AH
MOV CL, 3
SHR AL, CL

a) 51H b) 45H c) 50H d) 05H

4] The condition on which the instruction JBE takes place is:

a) CF= 0 and ZF= 0
b) CF= 1 and ZF= 1
c) CF= 1 or ZF= 0
d) CF= 1 or ZF= 1

5] Which instruction is used to convert ASCII to unpacked BCD?

a) ADD b) AND c) XOR d) CMP

6] Which instruction of the following use Register Addressing Mode?

a) MOV AL,29
b) MOV CX,[SI]
c) MOV AX,[200]
d) MOV CX,DX

7] Which instruction among the following would not change the content of register AL?

a) AND AL, 0FFH b) OR AL, 0FFH
c) XOR AL, 0FFH d) ADD AL, 0FFH

0011 1100 B, 3C

0001 0010

0100 1110

8] What would be the value of CF, SF, and PF after executing the following?

MOV BH, 3Ch
ADD BH, 12h

a) CF= 0, SF= 0, PF=0
b) CF= 0, SF=0 , PF=1
c) CF= 1, SF=0 , PF=1
d) CF= 0, SF=1 , PF=0

9] Which of the following defines Packed BCD number equals 18?

a) n1 db '18'
b) n2 db 18
c) n3 db 18h
d) n4 dw 0108h

Question#2

Replace each one of the following with an equivalent instruction **(3 points):**

1) NOT DH

Xor DH, FF

2) MOV AL, 0

SUB AL, AL

3) MOV DL, 2
DIV DL

MOV offset [DL], 2
DIV offset [DL]

Question#3

Given the following 2-byte short-jump instruction **(6 points):**

CS:IP	Instruction	Machine Code
2020H:1200H	JNZ Location	75XXH

a) Given ZF=0, What's the value of the XX that makes the code enters in an infinite-loop (stuck executing the JNZ Location instruction):

2020 : 1222
20200 + 1222 = 21422 = 3XX22

b) What's the value of XX that keeps the execution of the code sequentially regardless the value of the ZF?

XX = 00

c) If the value of XX is 32H, what makes the code executes the instruction located at address 2020H:1202H directly after the JNZ location instruction:

2020 : 1222 => 20200 + 1232 = 21432

d) If the JNZ instruction transfers the execution of the code to physical address 21445H, then the value of XX is:

2020 : 1245

20200
01245
21445

علامة النجاح 50 ex: 52

cmp AL, 50
JA

2

Question#4

a) Write a code to count the number of passing students? (3 points)

Grades db 40, 65, 82, 95, 72, 57, 62, 45, 35, 89
Count db ?

```
mark: move, AL, offest- Grades
move CL, 10
move BL, 0
cmp AL, 50
JA pass
move count, BL
pass: offest- Grades +1
inc BL
dec CL
JA mark
```

2

هنا لما ...

b) Write an assembly program to count the numbers that are multiples of 3 in the following array? (4 points)

X db 15, 20, 25, 36, 41, 12, 17
Count db ?